

A mathematical modeling approach to utilize CO₂ into hydrogen and electricity via Na-CO₂ electrochemical system

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Abstract

The increasing CO₂ concentration, now at a record 419.3 ppm, is causing environmental problems and current methods to reduce it are often inefficient and expensive¹. Electrochemical conversion of CO₂ in aqueous electrolytes into a useful product has the potential to reduce the anthropogenic CO₂ concentration. CO₂ gas dissolves in aqueous solutions and produces HCO₃⁻, H⁺, and CO₃²⁻ and continuous purging of CO₂ lowers the pH, increasing H⁺ production². Kim et al.³ have devised a hybrid Na-CO₂ system to utilize this to produce H⁺, hence CO₂ gas using a Na-metal as an anode, organic electrolyte, NASICON separator, aqueous electrolyte (CO₂ saturated NaOH solution) and Pt/C + IrO₂ coated carbon felt as the cathode. This system promises not only CO₂ utilization but also electricity and H₂ production.

This study develops a physics-based 2D multiphase transient mathematical model of the Na-CO₂ system using dilute solution theory, binary electrolyte theory, and laminar bubble flow in aqueous electrolyte. Using the simulated model H₂ production, local electrolyte current and potential distribution, species conception, and gas flow patterns are analyzed. The model can be used to optimize geometric configurations, flow rates, CO₂ conversion, and chemical reactions and can potentially promote environmental sustainability and make the world greener.

References

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